

ABDULLAHI MOHAMUD AHMED

Prospective Graduate Researcher · NLP & Low-Resource Language Technology

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Academic portfolio & research outputs: Preprint (Research Square) rs-10945954/v1 · Code github.com/Ogaal-Labs/somali_dictionary_lexicon · Dataset (Zenodo) [10.5281/zenodo.20509390](https://doi.org/10.5281/zenodo.20509390) · Portfolio abdallammud.github.io

RESEARCH INTERESTS

Low-resource language technology, with a focus on Somali natural language processing; computational lexicography and dataset construction; morphological analysis and part-of-speech tagging for morphologically rich languages; large language models and retrieval for under-resourced settings; and reproducible research with open data and software. I am motivated by building the foundational data resources and tools that under-resourced languages need in order to participate in modern NLP.

EDUCATION

BSc in Statistics and Planning · **SIMAD University**, Mogadishu

Sep 2015 – Dec 2019

- Faculty of Economics, Department of Statistics and Planning. Foundation in statistical analysis, probability, sampling and quantitative methods.
- Completed the university's concurrent English-language program (2019).

PUBLICATIONS & PREPRINTS

Ahmed, A. M., & Ugas, A. (2026). *Qaamuuska-NLP: A Structured 46K-Entry Somali Lexicon Extracted from a Print Dictionary*. Preprint, Research Square (rs-10945954/v1), CC-BY 4.0. **Under review at *Language Resources and Evaluation*** (Springer Nature). Produced with OgaalLabs. **First author.**

RESEARCH DATASETS & SOFTWARE

Ahmed, A. M. (2026). *Qaamuuska Af-Soomaaliga: Structured Somali NLP Dataset (46,314 entries)* [Data set]. Zenodo. DOI [10.5281/zenodo.20509390](https://doi.org/10.5281/zenodo.20509390). Code: github.com/Ogaal-Labs/somali_dictionary_lexicon (MIT).

Somali Morphological Analyzer (2026). Open-source suffix back-off part-of-speech and morphology model trained on the 46K-entry lexicon; distributed as a Python (pip) package with an interactive web demo.

Somali Data Story (2026). Interactive data visualisation (Chart.js) of the 46,314-entry dataset: letter frequency, word-length distribution and part-of-speech composition.

RESEARCH EXPERIENCE

Lead Researcher & First Author, Qaamuuska-NLP · **with OgaalLabs**

2026 – Present

- **Dataset construction:** Built a structured 46,314-entry Somali lexicon from a print dictionary using a Python pipeline (pdfplumber, regular expressions, JSON); designed the schema capturing headword, part of speech, homonym index, gender, verb class and transitivity, conjugation, singular/plural, domain, synonyms and definitions.
- **Contribution:** First Somali lexicon derived from a print dictionary; larger in distinct entries than prior 1,247-root and 5,584-root Somali lexicons, and richer in per-entry structure than lemma-plus-POS resources.
- **Quality evaluation:** Automated intrinsic checks over all 46,314 entries (0% POS parse failures, 95.4% cross-reference resolution, 0.19% duplicate keys) plus a reproducible random 100-entry manual spot-check against the source scans.
- **Tools & applications:** Built a full-text and prefix-search web dictionary and a suffix back-off morphological/POS analyzer; the resource supports POS tagging, lemmatisation, morphological analysis, MT-lexicon seeding and embedding training.

Research Collaborator · **OgaalLabs**

2026 – Present

- Member of a Somali language-technology research team; collaborators are graduate researchers based in South Korea, including the paper's co-author, currently a PhD candidate at a South Korean university.

TECHNICAL & RESEARCH SKILLS

- **Programming & NLP:** Python (Pandas, NumPy, pdfplumber, regex); applied NLP (tokenisation, morphological analysis, POS tagging); machine learning (regression, classification).
- **Data & databases:** SQL and relational design (MySQL, PostgreSQL, SQL Server, Oracle); large-scale dataset construction, cleaning, validation and deduplication.
- **Statistics:** Statistical analysis and quantitative methods; SPSS, R, Stata, advanced Excel (BSc in Statistics).
- **Tooling:** Git & GitHub, LaTeX, Chart.js, Power BI, Tableau; reproducible, version-controlled research pipelines.

SELECTED PROFESSIONAL EXPERIENCE

Director, Data & Software Department · **SOSTEC Technologies**, Mogadishu

Jan 2023 – Present

- Design and own databases, dashboards and full-stack data pipelines for enterprise, government and healthcare systems used across Somalia; lead a developer team on data quality, validation and reporting. (Software Developer, 2019–2022.)
- Deep hands-on engineering across large multi-module systems — relational schema design, transactions, role-based access control, API integrations and CI/CD.

CERTIFICATIONS

- Google Advanced Data Analytics Professional Certificate · Coursera, 2026
- Meta Database Engineering Professional Certificate · Coursera, 2026
- WorldQuant Data Science Lab · WorldQuant University, 2026
- Oracle Cloud Infrastructure (OCI) Developer & DevOps Professional · Oracle, 2025
- Data Analyst Nanodegree · Udacity, 2021

LANGUAGES

- **Somali:** Native; deep knowledge of Somali morphology and orthography developed through original lexicographic research.
- **English:** Fluent (professional working proficiency); language of research, documentation and publication.
- **Arabic:** Intermediate. **Korean:** Beginner, in progress (preparing for TOPIK).

REFERENCES

Ms. Zahra Abdikarim Ibrahim

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